Why do dynamic, nonlinear strategies yield false measures of alpha? Because buying and selling options—or any dynamic strategy—changes the distribution of returns.[18] Static measures, like the alpha, information, and Sharpe ratios, capture only certain components of the whole return distribution. Often, short volatility strategies can inflate alphas and information ratios because they increase negative skewness. These strategies increase losses in the left-hand tails and make the middle of the distribution "thicker" and appear to be more attractive to linear performance measures. Skewness and other higher moments are not taken into account by alphas and information ratios.

There are two ways to account for nonlinear payoffs.

### *Include Tradeable Nonlinear Factors*

Aggregate market volatility risk is an important factor, discussed in chapters 7 through 9, and an easy way to include the effects of short volatility strategies is to include volatility risk factors. Other nonlinear factors can also be included in factor benchmarks. By doing so, the asset owner is assuming that she can trade these nonlinear factors by herself. Sometimes, however, the only way to access these factors is through certain fund managers. In chapter 17, I will show how controlling for nonlinear factors crucially changes the alphas of hedge funds. Fung and Hsieh (2001), among many others, show that hedge fund returns often load significantly on option strategies.

### *Examine Nontradeable Nonlinearities*

It is easy to test whether fund returns exhibit nonlinear patterns by including nonlinear terms on the right-hand side of factor regressions. Common specifications include quadratic terms, like $r_t^2$, or option-like terms like $\max(r_t, 0)$.[19] The disadvantage is that, after including these terms, you do not have alpha—we always need tradeable factors on the right-hand side to compute alphas.

But we must move beyond alpha if we want evaluation measures that are robust to dynamic manipulation. These will not be alphas, but they can still be used to rank managers and evaluate skill. One state-of-the-art measure has been introduced by Goetzmann et al. (2007).[20] With long enough samples, this measure cannot be manipulated in the sense that selling options will not yield a false measure of performance.

The Goetzmann et al. evaluation measure is

$$\frac{1}{1-\gamma} \ln\left(\frac{1}{T}\sum_{t=1}^{T}(1+r_t-r_{ft})^{1-\gamma}\right), \tag{10.16}$$

[18] This is true even of simple rebalancing; see chapter 4.

[19] These can be traced to Treynor and Mazuy (1966) and Henriksson and Merton (1981), respectively.

[20] For some other notable manipulation-free performance measures, see Glosten and Jagannathan (1994) and Wang and Zhang (2011).